JNTUH USED 20-11-17PM Set 2

Code No: 117DU **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year I Semester Examinations, November/December - 2017
IMAGE PROCESSING AND PATTERN RECOGNITION
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

Part- A (25 Marks)

1.a)	What is meant by m- connectivity?	[2]
b)	What are different sources for degradation of image?	[3]
c)	Write a mask for gradient operation.	[2]
d)	Which morphological operation is equivalent of logic AND operation.	[3]
e)	What is fidelity in image processing?	[2]
f)	Compare statistical and spatial compression of image.	[3]
g)	What is meant by representation in an image?	[2]
h)	List the principle components of descriptors.	[3]
i)	What are the applications of pattern recognition?	[2]
j)	How are patterns classified based on likelihood function?	[3]

Part-B (50 Marks)

2. What is histogram? Explain the method of histogram equalization and how histogram is used to enhance the image? [10]

OR

3.a) Explain how sharpening low pass filter is used in image processing?
b) What are the limitations of inverse filter while using for image restoration? [5+5]

4. Explain the following morphological operations on an image:
a) Dilation b) Erosion c) Opening d) Closing e) Hit or miss [10]

OR

5. Explain the region based segmentation methods and also write the necessary conditions. [10]

6.a) How are image compression methods classified? Explain them.
b) Explain how contour coding is used for image compression? [5+5]

OR

7. Explain how predictive codes are used for both lossy and lossless image compression? [10]

8. How are skeltons used to represent image and explain with neat diagrams. [10]

OR

9. Why are descriptors used in image processing? Explain boundary descriptors. [10]

10.a) Explain basic concepts of pattern recognition with suitable examples.
b) Draw a simple pattern recognition model and explain it. [5+5]

OR

11. Write k-means algorithm and explain how it is used in pattern recognition. [10]

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